

# **TEST ANALYSIS REPORT**

## **Project Name:**

**Guru99\_Banking\_Automation**

## **GitHub Repository:**

[https://github.com/bharathkumar172002/Wipro\\_CapstoneProject](https://github.com/bharathkumar172002/Wipro_CapstoneProject)

### **1. OBJECTIVE**

The objective is to automate the functional testing of the Guru99 Bank application using Selenium WebDriver, TestNG, Maven, Jenkins, Page Object Model (POM), Extent Reports, and a Docker

### **1. FUNCTIONAL VALIDATION PERFORMED**

This section outlines the core banking operations that were successfully validated by the automation suite:

#### **1. Authentication Module**

- **Valid Login Verification:**
  - Ensures that a user with correct credentials gains successful access to the Manager Dashboard.
- **Invalid Login Validation:**
  - Confirms the system blocks unauthorized access and displays the appropriate error message ("User or Password is not valid").
- **Empty Login Validation:**
  - Verifies that the system triggers an alert when attempting to submit the form without any credentials.

#### **2. Customer Management Module**

- **New Customer Creation:**
  - Automates the entry of all required fields to register a new customer in the bank database.
- **Customer ID Validation:**
  - Ensures that the system generates a unique and valid Customer ID upon successful registration.

### 3. Account Management Module

- **New Account Creation:**
- Automates the process of opening a new account for a registered customer.
- **Account Details Verification:**
- Validates that the account information (Account Type, Initial Deposit, etc.) matches the input data.

### 4. Transaction Module

- **Deposit Transaction:**
- Confirms that depositing money updates the account balance accurately.
- **Withdrawal Transaction:**
- Verifies that withdrawing money correctly reduces the account balance.

### 5. Fund Transfer Module

- **Valid Fund Transfer:**
- Ensures that money can be successfully transferred from one account to another.
- **Invalid Account Validation:** Confirms the system triggers an error when an invalid target account ID is provided.
- **Insufficient Balance Validation:**
- Validates that the system prevents fund transfers or withdrawals if the account balance is lower than the required amount.

## 2. MODULE-WISE EXECUTION STATUS

| Module       | Status        |
|--------------|---------------|
| Login Module | 3/3<br>Passed |

|                               |                       |
|-------------------------------|-----------------------|
| <b>Customer Management</b>    | <b>1/1<br/>Passed</b> |
| <b>Account Management</b>     | <b>1/1<br/>Passed</b> |
| <b>Deposit Transaction</b>    | <b>2/2<br/>Passed</b> |
| <b>Withdrawal Transaction</b> | <b>2/2<br/>Passed</b> |
| <b>Fund Transfer</b>          | <b>3/3<br/>Passed</b> |

### 3. TEST ANALYSIS

- **Functional Testing: 100%**
- **Regression Testing: 100%**
- **UI Validation: 95%**
- **Smoke Testing: 100%**
- **Negative Testing: 100%**

### 4. CROSS BROWSER TESTING

- **Status:**
- **Chrome Browser Executed Successfully.**
- **Config:**
- **Edge Browser configuration is available and ready.**
- **Flexibility:**
- **Framework supports browser selection through the configuration file.**

### 5. TEST EXECUTION

| Metric              | Value |
|---------------------|-------|
| Total Test Cases    | 12    |
| Passed Test Cases   | 12    |
| Executed Test Cases | 12    |
| Failed Test Cases   | 0     |
| Pass Percentage     | 100%  |

## PERFORMANCE & EXECUTION ANALYTICS

### 6.1 Extent Report Insights

- **Execution Snapshot:**
- The framework's dashboard showcases a 100% success rate across all 12 automated test cases, with zero failures or skipped scenarios.
- **Feature Coverage:**
- The report rigorously validates core banking workflows, including:
  - **Authentication:**
    - Login Validation (Valid/Invalid/Empty credentials).
  - **Core Banking:**
    - Customer Registration & Account Management modules.
  - **Transaction Engine:**
    - Real-time validation for Deposits, Withdrawals, and Fund Transfers.
  - **Negative Scenarios:**

- Robust handling of edge cases such as "Invalid Account ID" and "Insufficient Balance" errors.

Jun 12, 2026 12:43:29 am

Tests

|                      |                            |      |
|----------------------|----------------------------|------|
| emptyLoginTest       | 12:43:35 am / 00:00:00:632 | Pass |
| invalidLoginTest     | 12:43:39 am / 00:00:01:159 | Pass |
| validLoginTest       | 12:43:43 am / 00:00:02:698 | Pass |
| createCustomerTest   | 12:43:48 am / 00:00:13:465 | Pass |
| validateCustomerId   | 12:44:02 am / 00:00:00:002 | Pass |
| createAccountTest    | 12:44:22 am / 00:00:07:029 | Pass |
| verifyAccountDetails | 12:44:29 am / 00:00:00:000 | Pass |
| depositMoneyTest     | 12:44:31 am / 00:00:04:381 | Pass |

emptyLoginTest

06.12.2026 12:43:35 am 06.12.2026 12:43:36 am 00:00:00:632 #test-id=1

| STATUS | TIMESTAMP   | DETAILS     |
|--------|-------------|-------------|
| Pass   | 12:43:36 am | Test Passed |

Jun 12, 2026 12:43:29 am

Tests

|                           |                            |      |
|---------------------------|----------------------------|------|
| createAccountTest         | 12:44:22 am / 00:00:07:029 | Pass |
| verifyAccountDetails      | 12:44:29 am / 00:00:00:000 | Pass |
| depositMoneyTest          | 12:44:31 am / 00:00:04:381 | Pass |
| withdrawalMoneyTest       | 12:44:38 am / 00:00:20:321 | Pass |
| fundTransferTest          | 12:45:01 am / 00:00:08:316 | Pass |
| invalidAccountDepositTest | 12:45:17 am / 00:00:10:952 | Pass |
| insufficientBalanceTest   | 12:45:30 am / 00:00:06:975 | Pass |

emptyLoginTest

06.12.2026 12:43:35 am 06.12.2026 12:43:36 am 00:00:00:632 #test-id=1

| STATUS | TIMESTAMP   | DETAILS     |
|--------|-------------|-------------|
| Pass   | 12:43:36 am | Test Passed |

Jun 12, 2026 12:43:29 am

Started Jun 12, 2026 12:43:29 am

Ended Jun 12, 2026 12:45:37 am

Tests Passed 12

Tests Failed 0

Tests

12 tests passed  
0 tests failed, 0 skipped, 0 others

Log events

12 events passed  
0 events failed, 0 others

Timeline



## Emailable report:

An Emailable Report is essentially a summary or a snapshot of your project's performance that is formatted to be sent via email.

In the context of software testing (like your Guru99 project), it is used to keep stakeholders, managers, or team members updated about the Test Results without them needing to log into your testing tools or run the code themselves.

 |  YouTube

 JioCinema - Watch...

 Gmail

 YouTube

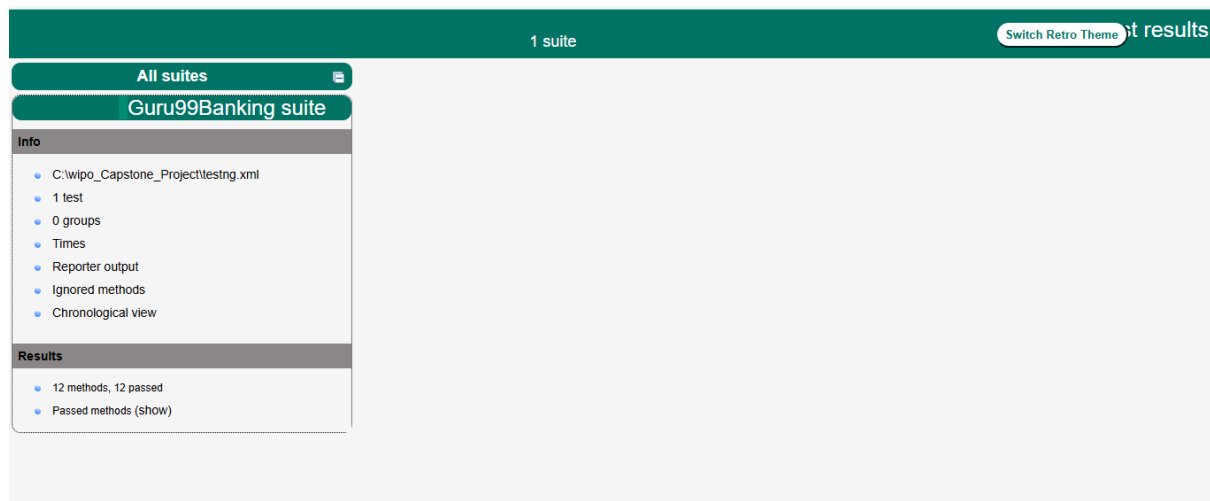
 Maps

| Test                            | # Passed | # Skipped | # Retried | # Failed | Time (ms) | Included Groups | Excluded Groups |
|---------------------------------|----------|-----------|-----------|----------|-----------|-----------------|-----------------|
| Guru99Banking suite             |          |           |           |          |           |                 |                 |
| Guru99_Banking_Automation Tests | 12       | 0         | 0         | 0        | 127,242   |                 |                 |

| Class                                    | Method                    | Start         | Time (ms) |
|------------------------------------------|---------------------------|---------------|-----------|
| Guru99Banking suite                      |                           |               |           |
| Guru99_Banking_Automation Tests — passed |                           |               |           |
| testcases.AccountDetailsTest             | verifyAccountDetails      | 1781205269497 | 0         |
| testcases.AccountTest                    | createAccountTest         | 1781205262464 | 7030      |
| testcases.CustomerTest                   | createCustomerTest        | 1781205228963 | 13465     |
| testcases.CustomerValidationTest         | validateCustomerId        | 1781205242441 | 3         |
| testcases.DepositTest                    | depositMoneyTest          | 1781205271663 | 4384      |
| testcases.FundTransferTest               | fundTransferTest          | 1781205301678 | 8317      |
| testcases.InsufficientBalanceTest        | insufficientBalanceTest   | 1781205330424 | 7000      |
| testcases.InvalidAccountTest             | invalidAccountDepositTest | 1781205317569 | 10950     |
| testcases.LoginTest                      | emptyLoginTest            | 1781205215487 | 641       |
|                                          | invalidLoginTest          | 1781205219095 | 1164      |
|                                          | validLoginTest            | 1781205223338 | 2698      |
| testcases.WithdrawalTest                 | withdrawalMoneyTest       | 1781205278084 | 20321     |

## TestNG Report Analysis:

The TestNG Report presents a comprehensive overview of the execution results for the Guru99 Bank Automation Test Suite. The report shows that all 12 test methods were executed successfully, with 12 tests passing, 0 failures, 0 skipped tests, and 0 retries, indicating smooth execution and complete validation of the application's functionality. The executed test cases cover key banking operations such as Login Validation, Customer Creation, Customer ID Verification, Account Creation, Account Details Validation, Deposit, Withdrawal, Fund Transfer, Invalid Account Handling, and Insufficient Balance Verification. Every test case is marked with a green pass status, confirming that the actual application behaviour matched the expected outcomes. The successful completion of all test scenarios demonstrates the robustness, reliability, and functional correctness of the Guru99 Bank application within the scope of the executed test suite.



## 7. TestNG Suite report

- **Method Reliability:**
- The TestNG execution engine confirmed a clean run of all 12 test methods with a flawless "Pass" status.
- **Functional Integrity:**
- The results demonstrate high robustness, confirming that the actual system behavior is perfectly aligned with the predefined functional requirements.
- **Execution Logs:**

- Every test method is accompanied by detailed timestamps, ensuring complete traceability and transparency of the testing lifecycle throughout the Guru99 Banking application.

## Guru99\_Banking\_Automation Tests

|                              |                              |
|------------------------------|------------------------------|
| Tests passed/Failed/Skipped: | 12/0/0                       |
| Started on:                  | Wed Jun 10 12:29:03 IST 2026 |
| Total time:                  | 168 seconds (168909 ms)      |
| Included groups:             |                              |
| Excluded groups:             |                              |

| PASSED TESTS                                                                    |              |           |                                            |          |
|---------------------------------------------------------------------------------|--------------|-----------|--------------------------------------------|----------|
| Test method                                                                     | Attribute(s) | Exception | Time (seconds)                             | Instance |
| <b>depositMoneyTest</b><br>Test class: testcases.DepositTest                    |              | 9         | testcases.DepositTest@361c294e             |          |
| <b>verifyAccountDetails</b><br>Test class: testcases.AccountDetailsTest         |              | 0         | testcases.AccountDetailsTest@4a8a60bc      |          |
| <b>invalidAccountDepositTest</b><br>Test class: testcases.InvalidAccountTest    |              | 7         | testcases.InvalidAccountTest@314b8f2d      |          |
| <b>insufficientBalanceTest</b><br>Test class: testcases.InsufficientBalanceTest |              | 7         | testcases.InsufficientBalanceTest@664a9613 |          |
| <b>validLoginTest</b><br>Test class: testcases.LoginTest                        |              | 8         | testcases.LoginTest@65e61854               |          |
| <b>fundTransferTest</b><br>Test class: testcases.FundTransferTest               |              | 7         | testcases.FundTransferTest@285d851a        |          |
| <b>validateCustomerId</b><br>Test class: testcases.CustomerValidationTest       |              | 0         | testcases.CustomerValidationTest@669513d8  |          |
| <b>withdrawalMoneyTest</b><br>Test class: testcases.WithdrawalTest              |              | 7         | testcases.WithdrawalTest@7859e786          |          |
| <b>invalidLoginTest</b><br>Test class: testcases.LoginTest                      |              | 1         | testcases.LoginTest@65e61854               |          |
| <b>emptyLoginTest</b><br>Test class: testcases.LoginTest                        |              | 2         | testcases.LoginTest@65e61854               |          |
| <b>createAccountTest</b><br>Test class: testcases.AccountTest                   |              | 19        | testcases.AccountTest@3a1d593e             |          |
| <b>createCustomerTest</b><br>Test class: testcases.CustomerTest                 |              | 22        | testcases.CustomerTest@2f7a7219            |          |

## Negative test report:

### Negative Tests

|                              |                              |
|------------------------------|------------------------------|
| Tests passed/Failed/Skipped: | 6/0/0                        |
| Started on:                  | Wed Jun 10 14:17:32 IST 2026 |
| Total time:                  | 60 seconds (60319 ms)        |
| Included groups:             |                              |
| Excluded groups:             |                              |

(Hover the method name to see the test class name)

| PASSED TESTS                                                                        |              |           |                                             |          |
|-------------------------------------------------------------------------------------|--------------|-----------|---------------------------------------------|----------|
| Test method                                                                         | Attribute(s) | Exception | Time (seconds)                              | Instance |
| <b>excessiveWithdrawalTest</b><br>Test class: testcases.FundTransferNegativeTest    |              | 2         | testcases.FundTransferNegativeTest@5ef8df1e |          |
| <b>invalidLoginTest</b><br>Test class: testcases.LoginTest                          |              | 1         | testcases.LoginTest@24a1c17f                |          |
| <b>insufficientBalanceTest</b><br>Test class: testcases.FundTransferNegativeTest    |              | 4         | testcases.FundTransferNegativeTest@5ef8df1e |          |
| <b>invalidDepositAccountTest</b><br>Test class: testcases.FundTransferNegativeTest  |              | 5         | testcases.FundTransferNegativeTest@5ef8df1e |          |
| <b>emptyLoginTest</b><br>Test class: testcases.LoginTest                            |              | 0         | testcases.LoginTest@24a1c17f                |          |
| <b>invalidAccountTransferTest</b><br>Test class: testcases.FundTransferNegativeTest |              | 8         | testcases.FundTransferNegativeTest@5ef8df1e |          |



